

TAREK MAHMOUD ABDELRAKY AHMED

Azure Data Engineer | ETL/ELT Pipelines | Data Warehousing | SQL | Python | Azure | ML Applications
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PROFESSIONAL SUMMARY

Azure Data Engineer and Communications & Information Engineering student with hands-on experience designing, building, and orchestrating end-to-end ETL/ELT pipelines, SQL data warehouses, dbt models, Airflow workflows, and analytics-ready star schemas. Skilled in Python, SQL, PostgreSQL, SQL Server, Azure Data Factory, Azure Databricks, Docker, dbt, Apache Airflow, Power BI, and Streamlit. Strong project portfolio covering batch data pipelines, medallion architecture, data quality testing, advanced OLAP analytics, and production-style ML applications with measurable results.

TECHNICAL SKILLS

Data Engineering	ETL/ELT Pipelines, Data Warehousing, Data Modeling, Star Schema, Medallion Architecture, Data Quality, Orchestration
Cloud & Databases	Azure Data Factory, Azure Databricks, Azure SQL, SQL Server, PostgreSQL, MySQL
Programming & Analytics	Python, SQL, Pandas, NumPy, Scikit-learn, Matplotlib, Java, C
BI, Tools & DevOps	Power BI, Streamlit, Docker, dbt, Apache Airflow, Git/GitHub, DBeaver, SSMS
ML & Data Science	Classification, Feature Engineering, TF-IDF, Logistic Regression, Model Evaluation

SELECTED PROJECTS

- GenderClassy - Name-Based Gender Prediction App** | Python, Streamlit, Pandas, Scikit-learn, TF-IDF, Logistic Regression | [GitHub](#) | [Live Demo](#)
- Designed and deployed a production-grade hybrid ML system combining dictionary-first lookup with character n-gram TF-IDF and Logistic Regression fallback, achieving 99.71% accuracy and 99.53% dictionary hit rate.
 - Engineered a complete data pipeline for a 4M+ row multilingual dataset, including multi-source ingestion, Unicode-aware normalization for English, Arabic, and accented names, weighted label aggregation, and confidence-based inference routing.
 - Built a full-featured Streamlit application supporting single-name prediction, bulk text prediction, CSV/Excel upload prediction, downloadable enriched outputs, and robust handling for null or invalid names.
- E-Commerce Data Pipeline** | PostgreSQL, Python, Docker, dbt, Apache Airflow, Pandas, SQLAlchemy | [GitHub](#)
- Built an end-to-end Olist e-commerce data pipeline loading raw CSV datasets into PostgreSQL and transforming them into staging, marts, fact tables, dimensions, and KPI models.
 - Modeled analytics-ready star schema layers including customers, products, sellers, order items, payments, reviews, daily sales, revenue by category, and revenue by seller.
 - Orchestrated raw ingestion, dbt transformations, and dbt tests with Airflow; containerized PostgreSQL, Airflow, and dbt workflows using Docker Compose for repeatable local execution.
- SQL Server Data Warehouse** | SQL Server, T-SQL, Medallion Architecture, Star Schema, Power BI | [GitHub](#)
- Designed a modern SQL Server data warehouse integrating ERP and CRM CSV sources across Bronze, Silver, and Gold medallion layers.
 - Implemented ETL scripts for ingestion, data cleansing, standardization, and business-ready modeling; delivered customer, product, and sales analytics tables optimized for Power BI reporting.
- Solid Waste Data Warehouse** | SQL Server, T-SQL, BULK INSERT, ROLLUP, CUBE, GROUPING SETS | [GitHub](#)
- Built a medallion-style warehouse for solid waste collection analytics with dim_date, dim_station, dim_truck, and fact_trips models using BULK INSERT for high-volume loading.
 - Authored advanced OLAP analytical queries using ROLLUP, CUBE, GROUPING SETS, and GROUPING() to surface waste volume trends, city-level KPIs, and fleet performance insights.
- Real-Time Car Telemetry Simulation** | Python, JSONL, IoT Data, Docker | [GitHub](#)
- Contributed to a DEPI team project simulating realistic multi-sensor car telemetry using a state-machine driving model and JSONL stream output for downstream pipeline consumption.

PROFESSIONAL EXPERIENCE

Azure Data Engineer Intern | Digital Egypt Pioneers Initiative (DEPI) | Cairo, Egypt | Nov 2025 - Present

- Engineered ETL pipelines and data workflows using Python and SQL to automate ingestion, transformation, and loading for analytics and reporting workloads.
- Applied Azure Data Factory and Azure Databricks in practical data engineering scenarios, building analytics-ready datasets with transformation logic, data quality checks, and reporting requirements.
- Practiced modern data engineering patterns including batch processing, orchestration, cloud-based workflows, and structured data modeling.

Machine Learning Trainee | National Telecommunication Institute (NTI) | Cairo, Egypt | Jul 2025 - Aug 2025

- Completed a 120-hour intensive ML program and built supervised classification projects including Breast Cancer detection, Titanic survival prediction, and customer segmentation models.
- Used Pandas, Scikit-learn, and Matplotlib for exploratory data analysis, preprocessing, feature engineering, cross-validation, model evaluation, and visualization.

Embedded Systems Trainee | IMT School (CoreTech) | Cairo, Egypt | Jul 2024 - Apr 2025

- Developed embedded applications using C and AVR microcontrollers; practiced hardware interfacing, RTOS fundamentals, debugging, testing, and system-level problem solving.

EDUCATION

B.Sc. Communications & Information Engineering (In Progress) | Helwan University - Faculty of Engineering | Cairo, Egypt | Sep 2022 - Present

- Relevant coursework: Data Structures & Algorithms, Database Systems, Signal Processing, Computer Networks, Software Engineering.

LEADERSHIP & VOLUNTEERING

HR Specialist | FEHU-CPC (Faculty of Engineering, Helwan University) | Cairo, Egypt | Sep 2025 - Dec 2025

- Coordinated student participation for a competitive problem-solving initiative and managed organizer-participant communication across multiple events.

Volunteer | Lekaa Association | Cairo, Egypt | Jul 2023 - Dec 2023

- Supported children's educational and community development programs; strengthened teamwork, facilitation, and organizational skills.

LANGUAGES

Arabic: Native | English: Fluent